

Yann Baglin-Bunod

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Electrical and Computer Engineering
University of California, San Diego

SUMMARY

1st Year Master's Electrical and Computer Engineering student specializing in Machine Learning and Data Science with graduate coursework in statistical learning, optimization, and probabilistic inference. Proficient in PyTorch; strong Python/C/C++; hands-on projects spanning NLP pipelines, computer vision, and on-device ML. Built ML prototypes and ran MLflow-tracked ablation studies; solid mathematical foundations in linear algebra, stochastic processes, and convex optimization.

EDUCATION

- **University of California, San Diego** *Jan 2026 – Mar 2027 (Expected)*
Masters in Machine Learning and Data Science San Diego, CA
 - Relevant Coursework: Statistical Learning (SVMs, kernel methods, Gaussian processes), Optimization & GPU Acceleration of Deep Learning, Linear Algebra & SVD, Probabilistic Graphical Models (MCMC, HMMs), Software Foundations in C++
- **University of California, San Diego** *Graduated: 2025*
B.S. in Electrical Engineering, GPA: 3.75 San Diego, CA
- **École Polytechnique, IP Paris** *Aug 2023 – Mar 2024*
Global Exchange Program - Advanced AI Coursework Paris, France
 - Studied Vision Transformers, Self-Supervised Learning, and Generative AI (Stable Diffusion, LoRA, RLHF).
 - Worked with NLP (Transformers, Large Language Models, Sequence Labeling).
 - Applied Reinforcement Learning to autonomous agent decision-making (DQN, MDPs, Bandit Machines).

EXPERIENCE

- **OpenAirlines** *Jun 2025 – Aug 2025*
Data Science Intern Toulouse, France
 - Defined a solution pipeline converting natural language queries into structured analyses on flight data
 - Built a working prototype enabling user testing with over 95% measured accuracy on evaluation tasks.
 - Used MLflow to track experiments, compare ablations, and communicate results.
 - Presented prototype demos and findings to product and engineering stakeholders.
- **Summer Research Internship Program UCSD** *May 2024 – Sep 2024*
Medical Swallow Analysis Project San Diego, CA
 - Applied deep learning to medical diagnostics; implemented data augmentation pipelines
 - Developed a CNN segmenter and SVM classifier for swallowing disorder detection; ran MLflow-tracked ablations.
 - Engineered an audio segmenter to preprocess swallowing sounds for classification.
 - Presented research results and live demos in lab meetings to interdisciplinary audiences.
- **Digital Health Technologies Lab** *Feb 2024 – May 2024*
Medical Chatbot Development San Diego, CA
 - Developed a multi-agent system optimizing ChatGPT's medical self-triage capabilities.
 - Delivered project demos and explained results to lab researchers and non-technical stakeholders.
- **UC San Diego - Instructional Assistant** *Jan 2023 – Dec 2023*
ECE 15: C Programming Tutor San Diego, CA
 - Tutored 140 students in C programming; received top feedback for teaching effectiveness.

PERSONAL PROJECTS

- **Local RAG Pipeline for Scientific Claim Verification** *Jan – Mar 2026*
ECE 226 Final Project | GitHub - <https://github.com/Yann-b-b>
 - Built a fully local Retrieval-Augmented Generation pipeline on Apple Silicon, replacing cloud LLM calls entirely.
 - Ingested and chunked PDF documents using FAISS for dense vector retrieval; embedded with all-MiniLM-L6-v2.
 - Used Phi-3 Mini as the local inference engine; achieved 60.0% accuracy on SciFact vs. 47.3% no-RAG baseline (+12.7pp).
 - Evaluated retrieval quality with Recall@2 = 80.4%; documented findings in an ACL-format paper and slide deck.
- **Snoring Detection Android App** *Oct 2025 – Present*
GitHub - https://github.com/Yann-b-b/Snoring_Detection_APP
 - Developed a real-time snore detection app using a CNN-based TensorFlow Lite model deployed on Android.
 - Optimized and profiled model runtime performance using Qualcomm AI Hub for on-device inference acceleration.
 - Integrated live audio streaming and inference visualization in Kotlin with TensorFlow Lite's Android API.

TECHNICAL SKILLS AND INTERESTS

Programming & Software Design: Python, C, C++17 (OOP, STL, design patterns, memory management, RAII), Assembly, Data Structures & Algorithms

Machine Learning: Deep Learning (CNNs, RNNs, Transformers), NLP, Reinforcement Learning, Computer Vision, Statistical Learning (SVMs, kernel methods, Gaussian processes, graphical models), Bayesian inference

ML Systems & Optimization: Hardware-aware DL optimization (CPU/GPU co-design), model quantization and on-device inference, RAG pipelines (FAISS, vector embeddings, local LLMs), multi-agent systems

Math Foundations: Linear algebra (SVD, pseudoinverse, low-rank approximation), stochastic processes, convex optimization, probabilistic inference; interest in Recommender Systems and Information Retrieval

Training & Evaluation: Data augmentation pipelines; ablation studies with MLflow; trainer configuration and performance tuning

Frameworks & Tools: PyTorch, Keras, Scikit-learn, NumPy, MLflow, CUDA, MATLAB, TensorFlow Lite

Systems & Development: Embedded Systems, Real-Time Operating Concepts, Debugging & Testing, Computer Architecture

Soft Skills: Communication, Collaboration, Analytical Problem-Solving